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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

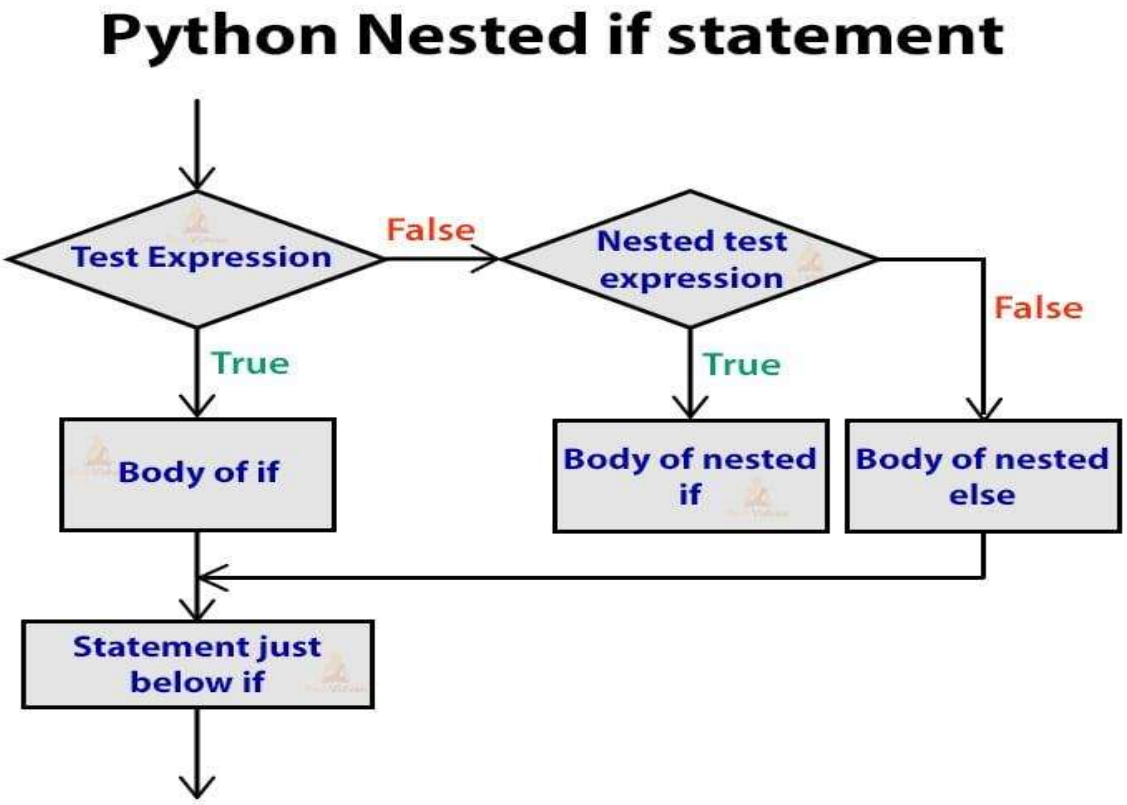
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 21: Nested If in Python



Learning Objectives

By the end of this lesson, students will be able to:

- ✅ Understand what a Nested if statement is.
- ✅ Learn how to use one if statement inside another.
- ✅ Solve real-life decision-making problems.
- ✅ Write Python programs using Nested if.
- ✅ Understand the importance of indentation.

What is a Nested If?

A **Nested if** means **an if statement inside another if statement**.

The **inner if** is checked **only if the outer if condition is True**.

Real-Life Example

Imagine you want to enter a library.

1. First, the librarian checks if you are a **student**.
2. If you are a student, they check if you have your **library ID card**.
3. If both conditions are true, you are allowed to enter.

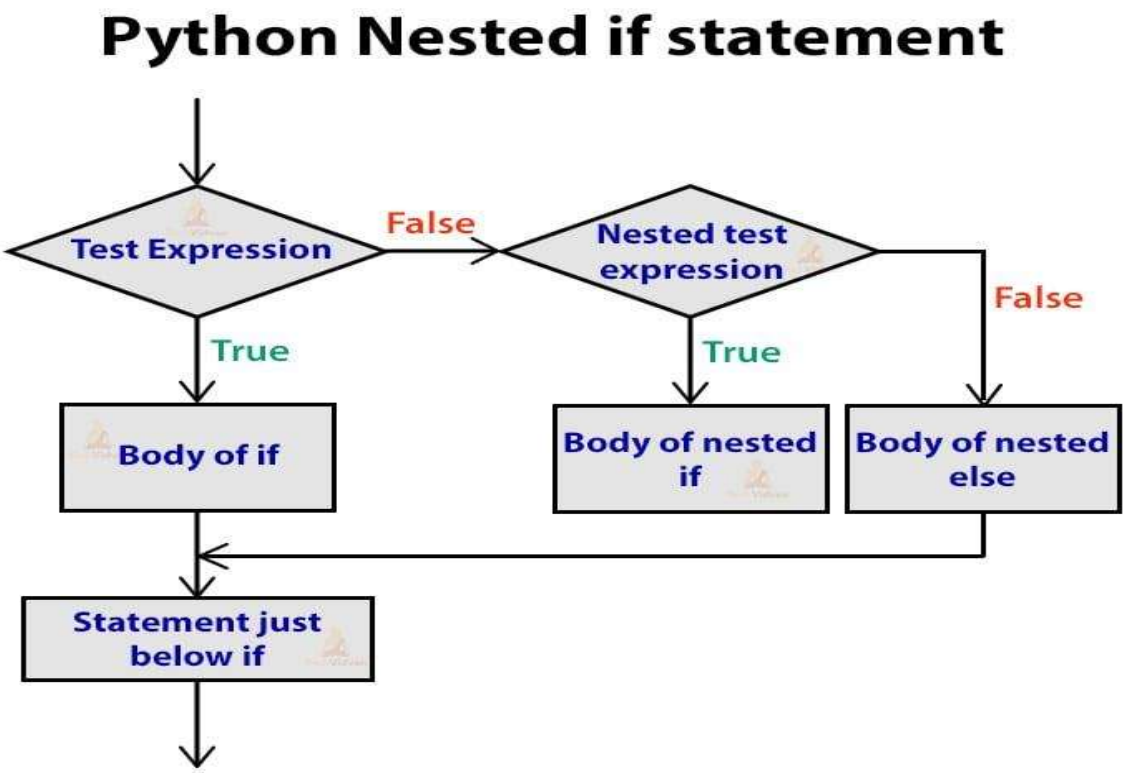
This is a **Nested If** because one decision is made inside another.

Nested If in Real Life





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Why Do We Use Nested If?

Nested if statements help us:

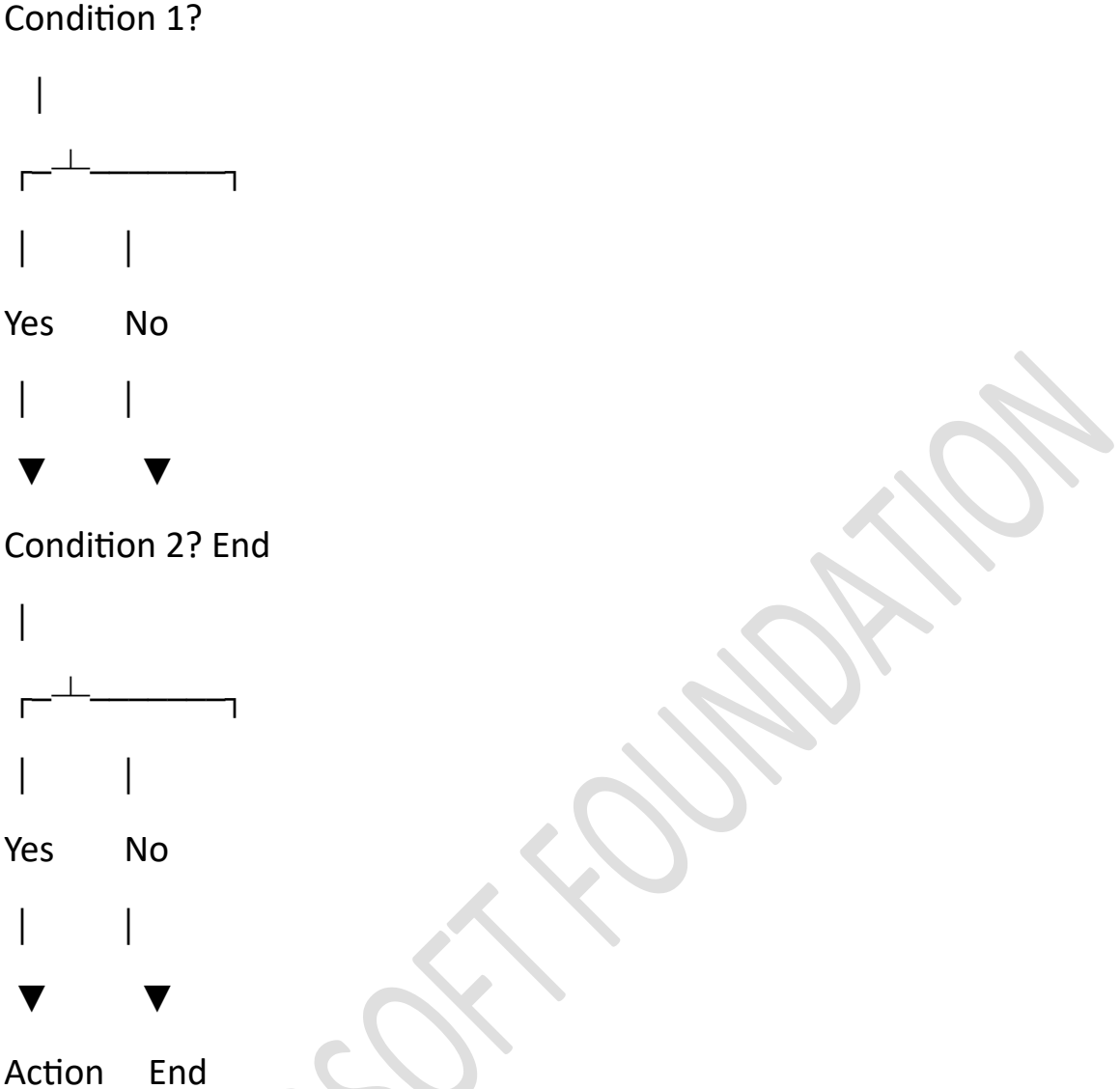
- Make decisions step by step.
- Check multiple conditions in order.
- Build secure login systems.
- Create games with different levels.
- Validate information before taking action.

Nested If Flow

Start

|

▼



Syntax of Nested If

```
if condition1:
   if condition2:
      statement
```

Example 1

age = 18

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```
if age >= 18:
    if age <= 60:
        print("You can participate.")
```

Output

You can participate.

Example 2

```
marks = 85

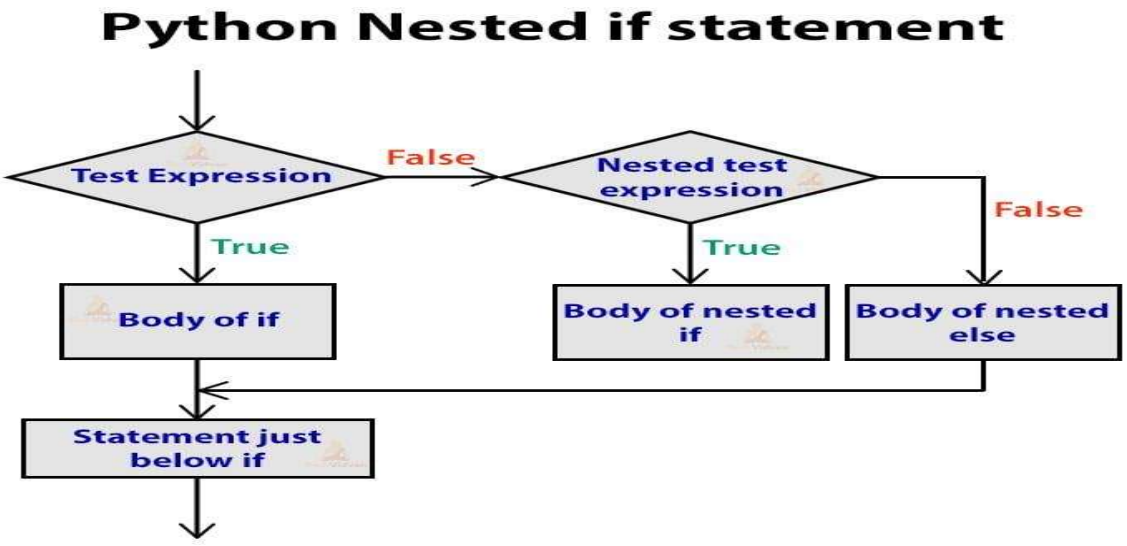
if marks >= 35:
    if marks >= 75:
        print("Excellent!")
```

Output

Excellent!

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Nested If Flowchart



```
age = 16
marks = 40

if age >= 18:
    if marks >= 80:
        print("Adult with high marks")
    else:
        print("Adult with low marks")
else:
    if marks >= 50:
        print("Minor with passing marks")
    else:
        print("Minor with failing marks")  # Nested else
```

```
Minor with failing marks
```



Understanding Nested If

age = 15

if age >= 10:

    print("Eligible for Junior Group")

```
    if age >= 15:
        print("Eligible for Senior Competition")
```

Output

```
Eligible for Junior Group
Eligible for Senior Competition
```

Example 3

```
username = "student"

password = "python123"

if username == "student":
    if password == "python123":
        print("Login Successful")
```

Output

```
Login Successful
```

Example 4

```
number = 12

if number > 0:
    if number % 2 == 0:
        print("Positive Even Number")
```

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Output

Positive Even Number

Example 5

```
temperature = 30


if temperature > 20:
    if temperature < 40:
        print("Pleasant Weather")
```

Output

Pleasant Weather

Indentation in Nested If

Python uses **indentation (spaces)** to show which code belongs to each if block.

Correct Example

```
age = 18


if age >= 18:
    print("Adult")


    if age >= 21:
        print("Can apply for special membership")
```

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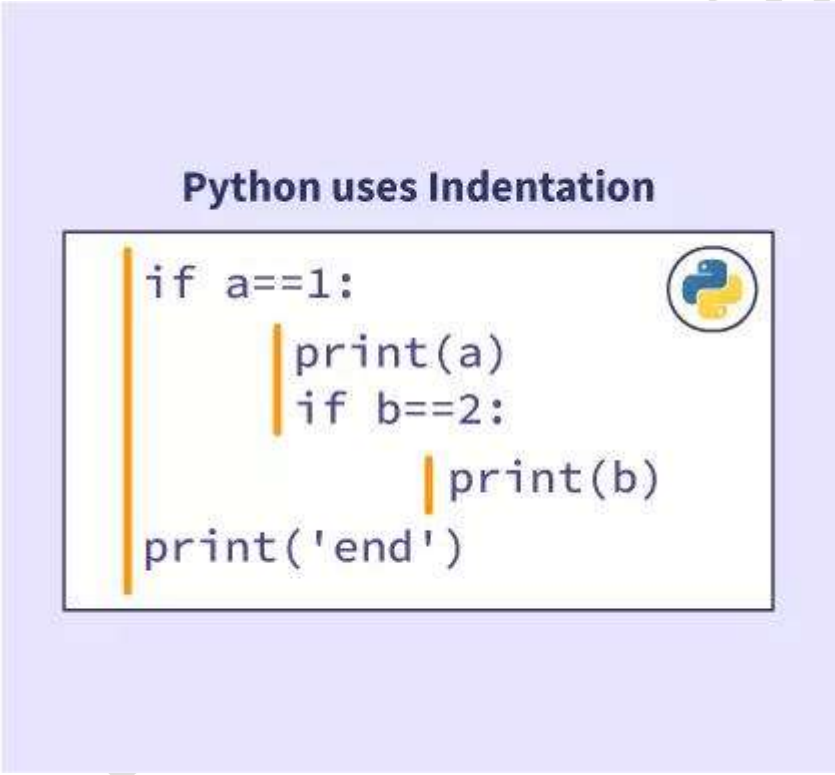
Wrong Example

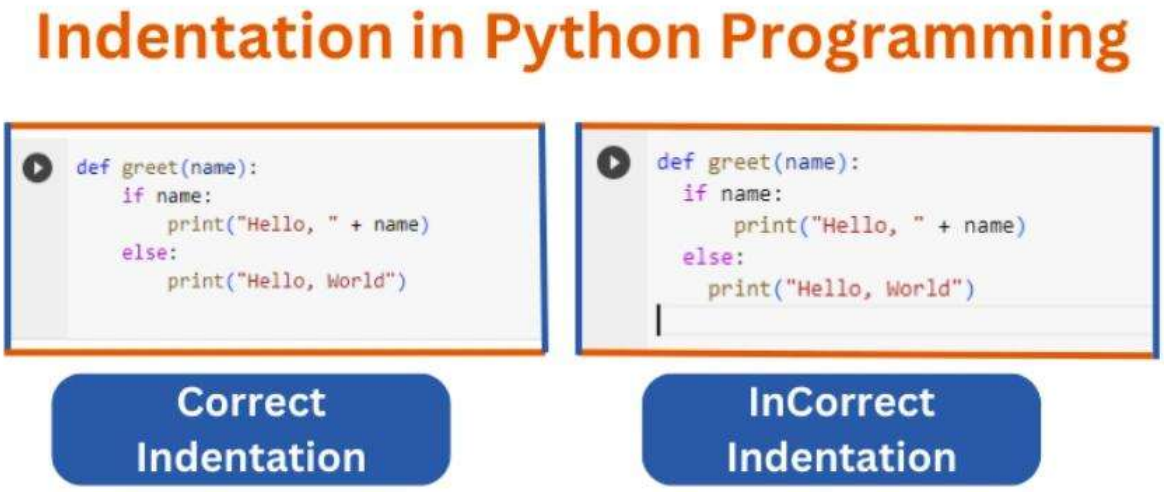
```
if age >= 18:
print("Adult")

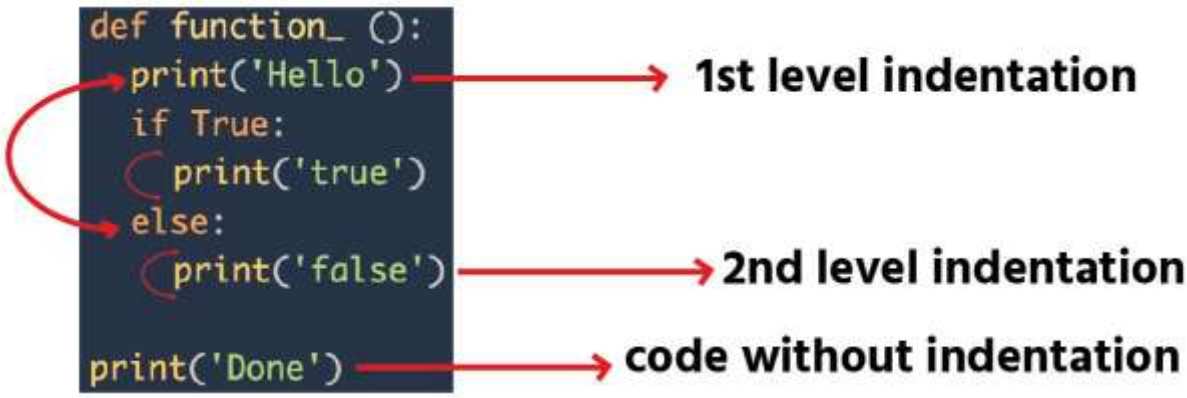

if age >= 21:
print("Can apply")
```

❌ This gives an **IndentationError**.

Correct vs Incorrect Indentation







Practical Session

Practical Activity 1

Library Entry

```
student = input("Are you a student? (yes/no): ")


if student == "yes":
    card = input("Do you have a library card? (yes/no): ")
```

```
    if card == "yes":
        print("Welcome to the Library!")
```

Practical Activity 2

Login Program

```
username = input("Enter Username: ")

if username == "admin":
    password = input("Enter Password: ")

    if password == "1234":
        print("Login Successful")
```

Practical Activity 3

Positive Even Number

```
number = int(input("Enter Number: "))

if number > 0:
    if number % 2 == 0:
        print("Positive Even Number")
```

Practical Activity 4

Student Grade

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```
marks = int(input("Enter Marks: "))

if marks >= 35:
    if marks >= 90:
        print("Grade A")
```

Practical Activity 5

Sports Team Selection

```
age = int(input("Enter Age: "))

if age >= 12:
    height = int(input("Enter Height (cm): "))

    if height >= 140:
        print("Selected for Sports Team")
```

Practical Activity 6

Movie Ticket

```
age = int(input("Enter Age: "))

if age >= 13:
    ticket = input("Do you have a ticket? (yes/no): ")

    if ticket == "yes":
```

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```
        print("Enjoy the Movie!")
```

Practical Activity 7

School Competition

```
class_name = int(input("Enter Class: "))

if class_name >= 7:
    permission = input("Permission Slip Submitted? (yes/no): ")

    if permission == "yes":
        print("You can participate.")
```

Practical Activity 8

ATM Withdrawal

```
pin = input("Enter PIN: ")

if pin == "1234":
    amount = int(input("Enter Amount: "))

    if amount <= 5000:
        print("Transaction Successful")
```

Classroom Coding Activity

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Activity 1

Write a program to check if a number is **positive and even**.

Activity 2

Create a login program using username and password.

Activity 3

Write a program to check whether a student can participate in a competition.

Activity 4

Write a program for ATM withdrawal with PIN verification.

Real-Life Applications

Situation	Nested If Use
ATM Machine	Check PIN → Check Balance
School Admission	Check Age → Check Documents
Library Entry	Check Student → Check ID Card
Online Login	Check Username → Check Password
Airport Security	Check Ticket → Check Passport
Sports Selection	Check Age → Check Height

Common Mistakes

❌ **Missing Indentation**

Wrong

```
if age >= 18:
```

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```
print("Adult")
```

✅ **Correct**

```
if age >= 18:
    print("Adult")
```

❌ **Using = Instead of ==**

Wrong

```
if pin = "1234":
```

✅ **Correct**

```
if pin == "1234":
```

❌ **Forgetting the Colon (:)**

Wrong

```
if age >= 18
```

✅ **Correct**

```
if age >= 18:
```

Best Practices

- ✓ Use meaningful variable names (age, marks, password).
- ✓ Keep indentation consistent (4 spaces is recommended).
- ✓ Write simple and readable conditions.
- ✓ Test your program with different inputs.
- ✓ Add comments to explain important parts of the code.

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Keywords

- **Nested If** – An if statement inside another if.
- **Condition** – A True or False expression.
- **Indentation** – Spaces that define a block of code.
- **Boolean** – A value that is either True or False.
- **Decision Making** – Choosing an action based on conditions.

Lesson Summary

- A Nested if is an if statement inside another if.
- The inner if runs only when the outer if condition is True.
- Nested if statements help solve problems that require multiple checks.
- Correct indentation is essential in Python.
- Nested if is commonly used in login systems, ATM machines, school admissions, and many other real-life applications.

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Quiz

Multiple Choice Questions

1. What is a Nested if?

A) A loop

B) An if statement inside another if statement

C) A function

D) A variable

✅ **Answer: B**

2. The inner if statement runs only when:

A) The program starts

B) The outer if condition is True

C) The condition is False

D) The program ends

✅ **Answer: B**

3. Which symbol is used to compare two values?

A) =

B) ==

C) +

D) *

✅ **Answer: B**

4. What happens if indentation is incorrect in Python?

A) The program becomes faster.

B) Python ignores it.

C) Python shows an **IndentationError**.

D) Nothing happens.

✅ **Answer: C**

5. Which real-life example best represents a Nested if?

A) Adding two numbers

B) Checking username first, then password

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C) Printing a message

D) Multiplying two numbers

✅ **Answer: B**

Homework

1. Write a program to check if a number is **positive and even**.
2. Create a login program using a username and password.
3. Write a program to check if a student is eligible for a sports competition based on age and height.
4. Create an ATM program that checks the PIN and withdrawal amount.
5. Write a program that checks whether a student has passed and then awards **Grade A** if the marks are **90 or above**.

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